Krishnaa **Vadivel** | Machine Learning Engineer | High-Performance Computing

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Profile

PhD researcher and engineer building machine-learning models and running them on high-performance computing systems for large-scale materials simulation. Combines model development, GPU-aware implementation, distributed compute, and scientific domain knowledge across materials, optics, and agentic ML tooling.

Research Experience

Yale University, Electrical Engineering
Research Assistant / PhD Researcher *2021–present*

- Developed equivariant graph neural networks in PyTorch and PyTorch Geometric, similar to DeepH and MACE, to accelerate optical-properties calculations and material-energy prediction with automatic differentiation.
- Built distributed compute and data-processing software in Python and C++ using MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc across national HPC environments including Frontier, Frontera, and Perlmutter.
- Connected ML methods to domain theory through first-principles exciton calculations, many-body modeling, and numerical simulation, enabling realistic ML problem framing for materials systems.
- Used agentic ML and retrieval-driven tooling to generate first-principles software inputs from journal articles and technical documentation.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state problems.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*

- Built CUDA-accelerated data-processing and visualization software for large waveform datasets, strengthening experience in GPU-enabled high-throughput pipelines.
- Contributed ASP-style services and Microsoft SQL-backed data systems for engineering operations and tooling.
- Worked close to embedded and instrumentation contexts, improving practical understanding of data generation, system integration, and performance constraints.

Selected Projects

Materials ML for Optical Properties

- Developed equivariant graph neural networks in PyTorch and torch_geometric to accelerate optical-properties calculations and material-energy prediction in materials systems.

Matrix-Multiply AI Accelerator

- Built a Verilog-based matrix-multiply accelerator project to demonstrate hardware-aware AI compute and datapath design fundamentals.

Agentic RAG for Scientific Workflows

- Built retrieval-driven tooling that reads software documentation and academic papers to generate structured starter inputs for computational science and quantum-chemistry-style tasks.

Distributed Scientific ML Infrastructure

- Combined HPC simulation infrastructure with CUDA/ROCm acceleration and scalable Python/C++ tooling to support ML-enabled scientific workloads.

Education

Yale University
PhD, Electrical Engineering *2021–present*

Ongoing work spanning scientific computing, first-principles materials simulation, and ML-assisted acceleration.

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Selected Coursework

Machine Learning: Probabilistic Machine Learning
Theory: Theoretical Computer Science; strong mathematical/theory preparation supporting modern ML work
Scientific Foundations: Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics
Engineering Foundations: Signals and systems, embedded systems, and numerical/scientific computing practice

Technical Skills

ML: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks, transformers, automatic differentiation
Languages: Python, C++, CUDA, JavaScript, Bash, PowerShell
Scientific Python: NumPy, pandas, SciPy, SymPy, matplotlib
Compute: MPI, OpenMP, CUDA, ROCm, PETSc, SLEPc, mpi4py, petsc4py, slepc4py
Platforms: Linux, Docker, Docker Compose, Kubernetes, gcloud, Frontier, Frontera, Perlmutter